

Current Treatment Strategies for Acromegaly: An Evidence-Based Review

DeepSeek + PubMed RAG Literature Synthesis Report

1. Research Question and Search Strategy

Research question:

DeepSeek-optimized PubMed query: acromegaly[MeSH] AND treatment[Title/Abstract]

English semantic translation: Current treatment strategies for acromegaly

Candidate PubMed articles: 18

2. Executive Summary

- Acromegaly is a chronic systemic disease caused by excessive growth hormone (GH) secretion, typically from a pituitary adenoma, and is associated with increased morbidity and mortality if not adequately treated [PMID:25248589, PMID:30184064].
- The three main treatment modalities are transsphenoidal surgery, medical therapy, and radiotherapy, with surgery considered the first-line approach achieving remission in approximately 50–55% of patients [PMID:40644375, PMID:15539805].
- Medical therapy includes somatostatin analogs (e.g., octreotide, lanreotide, pasireotide) as the cornerstone, the GH receptor antagonist pegvisomant, and dopamine agonists, used as adjuvant or primary treatment in selected cases [PMID:39586277, PMID:25792221].
- Persistent or recurrent disease after surgery requires additional interventions, including repeat surgery, medical therapy, or radiotherapy, to achieve biochemical control and reduce long-term complications [PMID:21167718, PMID:35688725].
- Effective treatment aims to normalize GH and insulin-like growth factor 1 (IGF-1) levels, which is crucial for restoring mortality rates to those of the general population and managing systemic complications such as hypertension and cardiovascular disease [PMID:31655776, PMID:30184064].

3. Introduction and Disease Overview

Acromegaly is a rare disorder with an estimated prevalence of 40–130 cases per million inhabitants, characterized by slowly progressive somatic disfigurement and systemic manifestations [PMID:25248589]. The condition results from GH hypersecretion, usually from a pituitary adenoma, and leads to elevated IGF-1 levels. Without adequate treatment, acromegaly is associated with a mortality rate more than twice that of the general population and significant comorbidities, including cardiovascular, respiratory, and metabolic complications [PMID:25248589, PMID:30184064]. The primary goal of treatment is to achieve biochemical remission, defined as serum GH levels below 2 µg/L and normal IGF-1 levels, which has been shown to reduce mortality and morbidity [PMID:15539805].

Evidence links: PMID:25248589PMID:30184064PMID:15539805

4. Surgical Management

Transsphenoidal adenoma resection remains the first-line treatment for acromegaly. It achieves remission targets in an average of 55% of patients, though persistence of GH hypersecretion occurs in approximately 43% of cases and recurrence in 2–3% [PMID:15539805, PMID:21167718]. Surgery is most effective for microadenomas and well-circumscribed macroadenomas. For patients not in remission postoperatively, options include repeat surgery or adjuvant therapy with medical treatment or radiotherapy [PMID:15539805, PMID:40644375]. Early surgical intervention is critical to prevent irreversible complications and improve long-term outcomes [PMID:25248589].

Evidence links: PMID:15539805PMID:21167718PMID:40644375PMID:25248589

5. Medical Therapy

Medical therapy plays a central role in acromegaly management, particularly for patients who are not cured by surgery or who are poor surgical candidates. The main classes of drugs include: (1) Somatostatin analogs (SSAs) such as octreotide, lanreotide, and pasireotide, which are the cornerstone of medical treatment and can be used as primary therapy in selected patients [PMID:39586277, PMID:25792221]. Long-acting formulations allow for convenient dosing and sustained biochemical control. (2) The GH receptor antagonist pegvisomant, which is the most recent addition and is highly effective in normalizing IGF-1 levels, even in patients resistant to SSAs PMID:25792221. (3) Dopamine agonists, though less potent, may be used in mild cases or as combination therapy. Pharmacoeconomic analyses, such as a Brazilian cost-utility study, have compared strategies involving these agents to guide treatment decisions PMID:32299084. The choice of agent depends on tumor characteristics, patient preference, and side effect profiles PMID:40644375.

Evidence links: PMID:39586277PMID:25792221PMID:32299084PMID:40644375

6. Radiotherapy and Combination Approaches

Radiotherapy is reserved for patients with persistent disease activity despite surgery and medical therapy. Fractionated external beam radiotherapy and stereotactic radiosurgery can achieve long-term tumor control and biochemical remission, though effects may take years to manifest [PMID:15539805, PMID:40644375]. It is often used in combination with medical therapy to bridge the period until radiation effects become apparent. The current treatment landscape emphasizes a multimodal approach, integrating surgery, medical therapy, and radiotherapy to address both tumor mass and hormonal hypersecretion [PMID:30184064, PMID:35688725].

Evidence links: PMID:15539805PMID:40644375PMID:30184064PMID:35688725

7. Management of Systemic Complications

Acromegaly is associated with numerous systemic complications, including hypertension (median frequency 33.6%), cardiovascular disease, diabetes mellitus, and sleep apnea syndrome [PMID:31655776, PMID:30184064]. The pathogenesis of hypertension involves expanded extracellular fluid volume, increased peripheral vascular resistance, and sleep apnea. Treatment of acromegaly can improve some complications, though the effect of GH/IGF-1 normalization on blood pressure remains unclear PMID:31655776. Systemic inflammation has also been observed in treated patients, and its association with vascular function is under investigation PMID:33964727. Comprehensive management requires addressing these comorbidities alongside GH excess PMID:30184064.

Evidence links: PMID:31655776PMID:30184064PMID:33964727

8. Latest Clinical / Research Progress

- Advances in medical therapy, including the development of long-acting somatostatin analogs and pegvisomant, have expanded treatment options and improved biochemical control rates [PMID:39586277, PMID:25792221].
- Innovative therapeutics are moving away from a trial-and-error approach toward personalized medicine, with new drugs and combination strategies under investigation PMID:35688725.
- Improved management of systemic complications, such as diabetes and heart failure, has contributed to reduced mortality in acromegaly patients over recent decades PMID:30184064.

9. Future Research Directions

- Development of novel pharmacological agents targeting GH and IGF-1 pathways with improved efficacy and safety profiles [PMID:35688725, PMID:33783289].
- Refinement of personalized treatment algorithms based on tumor biology, patient characteristics, and biomarkers to optimize outcomes PMID:40644375.

- Further research into the impact of treatment on systemic inflammation and cardiovascular risk to guide comprehensive care PMID:33964727.
- Exploration of cost-effective strategies and value of information analyses to inform healthcare policy, especially in resource-limited settings PMID:32299084.

10. Limitations

- The review is based on a limited set of PubMed articles and may not capture all recent developments or guidelines.
- Some provided abstracts lack full details on study design, patient populations, and long-term outcomes, limiting the depth of analysis.
- The evidence on newer pharmacological agents is still emerging, and their long-term efficacy and safety require further validation PMID:33783289.

Appendix: PubMed Reference List

PMID	Year	Journal	Title
31655776		Endocrinol Metab Clin North Am	Hypertension and Acromegaly.
25248589		Handb Clin Neurol	Acromegaly.
40644375		Endocr Rev	Medical Treatment of Acromegaly: Navigating the Present, Shaping the Future.
38991906		Best Pract Res Clin Endocrinol Metab	State-of-the-art practice and research, clinical endocrinology and metabolism preface to the special issue: Diagnosis and treatment of acromegaly.
33783289		Expert Opin Pharmacother	New and emerging pharmacological treatment options for acromegaly.
30184064		Endocr Rev	Systemic Complications of Acromegaly and the Impact of the Current Treatment Landscape: An Update.
15539805		Horm Res	Treatment of acromegaly.
33964727		Growth Horm IGF Res	The association between treatment and systemic inflammation in acromegaly.
21167718		J Clin Neurosci	Treatment of persistent and recurrent acromegaly.
11070991		Tidsskr Nor Laegeforen	[Acromegaly--diagnosis and treatment].
35688725		Best Pract Res Clin Endocrinol Metab	Novel therapeutic approaches in acromegaly.
32299084		Neuroendocrinology	Pharmacological Acromegaly Treatment: Cost-Utility and Value of Information Analysis.
39586277		Front Horm Res	Treatment with Long-Acting Somatostatin Analogues in Patients with Acromegaly: When and How.
25792221		Neuroendocrinology	Pegvisomant Treatment in Acromegaly.
38638137		Front Endocrinol (Lausanne)	Changes in acromegaly comorbidities, treatment, and outcome over three decades: a nationwide cohort study.
10770989		N Engl J Med	Treatment of acromegaly.
30843181		Endokrynol Pol	Diagnostics and treatment of acromegaly - updated recommendations of the Polish Society of Endocrinology.
26486485		Vnitr Lek	[Acromegaly: current view].

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